

# Akshitha Ganga B R

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## CAREER OBJECTIVE

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B.Tech ECE final-year student (2023–2027) seeking an internship to apply skills in embedded systems, Python, Java, C, and web development. Experienced in building real-world projects and leading technical events.

## EDUCATION

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### B.Tech — Electronics and Communication Engineering

2023 – 2027

Ananthalakshmi Institute of Technology and Sciences, Anantapur

CGPA: 8 / 10

## TECHNICAL SKILLS

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**Programming Languages:** Python, Java, C, HTML5, CSS3, JavaScript (ES6)

**Embedded Systems:** Arduino IDE, Sensor Integration, Microcontroller Programming, Signal Processing

**Electronics:** VLSI Design, Circuit Analysis, ECE Fundamentals

**HDL & Digital Design:** Verilog, RTL Design, Pipelining, Processor Architecture, Icarus Verilog

**Tools & Platforms:** VS Code, Git, GitHub Pages, Chrome DevTools, LinkedIn Learning

**Soft Skills:** Team Leadership, Event Coordination, Technical Presentations, Community Engagement

## PROJECTS

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### 3-Stage Pipelined RISC-V Processor (Verilog)

2025

- Designed and simulated a 3-stage pipelined RISC-V processor in Verilog — implementing Fetch, Decode, and Execute stages with ALU, Register File, and Control Unit; verified via testbench simulation with Icarus Verilog.
- Technologies: Verilog, Icarus Verilog, RISC-V ISA. GitHub: <https://github.com/akshithaganga07/riscv-processor>

### Smart Door Knock Recognition System

May 2025

- Designed and built a knock-pattern recognition system using Arduino IDE to enable secure, automated door access — reducing unauthorized entry risk by enforcing custom knock sequences.
- Integrated piezoelectric sensors and custom C/C++ algorithms to detect and validate specific knock patterns with high accuracy, achieving reliable real-time response.
- Applied embedded systems principles including signal processing and microcontroller programming, delivering a low-cost home security solution.
- Technologies: Arduino IDE, C/C++, Piezoelectric sensor modules, Serial communication.

### Personal Portfolio Website

2026

- Designed and developed a fully responsive personal portfolio using HTML5, CSS3 (Flexbox/Grid), and JavaScript — deployed live on GitHub Pages.
- Built interactive UI features including dynamic navigation, smooth scrolling, and contact form validation using vanilla JavaScript (ES6), improving user engagement.
- Ensured cross-device compatibility across desktop and mobile browsers through responsive design techniques and Chrome DevTools testing.
- Technologies: HTML5, CSS3, JavaScript (ES6), GitHub Pages. Live at: <https://github.com/akshithaganga07>

### CPU Scheduling Visualizer

2025

- Developed a web-based CPU scheduling simulator supporting FCFS, SJF, Round Robin, and Priority algorithms — visualizing process execution via animated Gantt charts and performance metrics.
- Built a Python Flask REST API backend implementing all four scheduling algorithms from scratch, computing waiting time, turnaround time, CPU utilization, and throughput in real time.
- Designed an interactive dark-themed dashboard using HTML5, CSS3, and JavaScript with Chart.js for stacked bar chart comparisons — directly applying NPTEL Computer Architecture concepts.

- Technologies: Python, Flask, HTML5, CSS3, JavaScript (ES6), Chart.js. GitHub: <https://github.com/akshithaganga07/cpu-scheduling-visualizer>

## CERTIFICATIONS & TRAINING

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### Computer Architecture and Organisation

*NPTEL (IIT) | 2025–2026*

Completed the NPTEL certification exam on Computer Architecture and Organisation — covering processor design, memory hierarchy, instruction sets, and pipelining.

### Data Analytics Job Simulation — Deloitte (via Forage)

*Deloitte x Forage | May 2026*

Completed Deloitte's virtual job simulation covering data analysis and forensic technology — gaining hands-on industry exposure from one of the Big 4 firms. Issued by Forage with verified certificate.

### Java, Python & C Programming

*LinkedIn Learning | 2025–2026*

Completed structured courses covering OOP concepts, data structures, algorithms, and core language fundamentals across three languages.

### Web Development Fundamentals (HTML, CSS & JavaScript)

*Self-Study & Online Resources | 2025–2026*

Mastered HTML document structure, CSS responsive layouts, and JavaScript DOM manipulation — applied directly in personal portfolio project.

## EXTRACURRICULAR ACTIVITIES

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- Served as Technical Event Coordinator at the annual college fest — managed logistics, participant registration, and live event flow for 100+ attendees.
- Represented the college at TechSpark Fest, Tadipatri Engineering College, delivering a technical presentation on an emerging ECE topic to a cross-institutional audience.
- Organized and won recognition in the college's internal PowerPoint Presentation competition, demonstrating research depth and communication clarity.
- Led community outreach initiatives promoting healthcare awareness and hygiene education in rural villages, coordinating a cross-functional team of volunteers.